

Lab 4: VibeCoding — Connect Four vs AI

Software Engineering Laboratory Submission — Task 2 LLM Chat History

STUDENT REPOSITORY	TARGET FOLDER	AI CODING ASSISTANT	CURRENT MILESTONE
abhiraamadiga/SE_submissions	Lab-4	Antigravity IDE (Gemini)	Task 2 — Complete Game Termination

Milestone Progress

Task	Requirement	File	Status
Task 1	Complete Win Detection (Diagonal Wins)	board.py	✓ Committed
Task 2	Game Termination, Draws, & Input Validation	game.py	Completed & Ready
Task 3	Intelligent AI Decision-Making	ai.py	Pending
Task 4	Move-Level Feedback	game.py	Pending

Prompt and Response History

User Prompt 1

"C:\Users\abhi1\Downloads\Lab_4_VibeCoding_Student_handout (1).pdf"

this contains instructions i want u to rename 15_connect_four to Lab-4 and also wait until i tell u to make changes and add this repo as remote git remote add origin https://github.com/abhiraamadiga/SE_submissions.git

dont do any changes in the 15_connect_four

dont push or add anything

AI Assistant Response

I have completed the initial setup as requested:

- Renamed directory:** 15_connect_four has been renamed to Lab-4 .
- Configured remote:** The remote origin is set to https://github.com/abhiraamadiga/SE_submissions.git .
- No changes made to code:** The files inside Lab-4 remain untouched.
- No git add / git push:** Nothing has been staged or pushed.

User Prompt 2

i made 10-seconds video whts the nexr step?

AI Assistant Response

Great job! Now that you have recorded the "before" video, the next step is to implement and commit the 4 tasks one by one starting with **Task 1** (fixing diagonal win detection in board.py).

User Prompt 3

Complete task 1

AI Assistant Response

Implemented and verified **Task 1 — Complete win detection** in `board.py` :

- Added diagonal directions `(1, 1)` and `(1, -1)` to `Board.winner()` .
- Verified with unit tests that all 4 directions detect wins and sequences `< 4` do not count.

User Prompt 4

complete task-2

AI Assistant Response

I have implemented and verified **Task 2 — Complete game termination** in `game.py` !

1. Defect Analysis & Requirements

- **Input Handling:** Non-numeric input, out-of-range numbers (< 1 or > 7), and empty inputs must not crash the game or alter board state.
- **Terminal Signals:** `Ctrl+C` and `EOFError` (e.g. `Ctrl+Z`) must exit cleanly rather than throwing an unhandled exception traceback.
- **Full Column Prevention:** Full columns must be rejected with clear feedback before modifying the board.
- **Immediate Game Termination:** A winning move or draw must terminate the game immediately before any further turns are requested.

2. Code Modifications (`game.py`)

```
# Catch exit signals cleanly
try:
    raw = input("Column (1-7), or q: ").strip().lower()
except (EOFError, KeyboardInterrupt):
    print("\nGame exited.")
    return

if raw == "q":
    print("Game quit.")
    return

# Validate numeric range
try:
    col_num = int(raw)
except ValueError:
    print("Invalid input. Enter a column number (1-7).")
    continue

if not 1 <= col_num <= 7:
    print("Column out of range. Enter a number between 1 and 7.")
    continue

col = col_num - 1
if self.board.is_column_full(col):
    print(f"Column {col_num} is full. Choose another column.")
    continue

# Immediate game termination
if self.board.winner(self.turn):
    self.board.print()
    print(self.turn, "wins!")
    return
if self.board.full():
    self.board.print()
    print("Draw.")
    return
```

3. Verification & Testing

- ✓ **Invalid string inputs:** Handled cleanly without altering the board.
- ✓ **Out-of-range numbers (0, 8, -2):** Prompted with range guidance without board change.
- ✓ **Full column check:** Rejected and prompts player again.

- ✓ **Terminal signals (EOF / Ctrl+C):** Exits cleanly without stack trace.
- ✓ **Win / Draw stopping:** Game halts immediately upon terminal board state.

Done when criteria satisfied: No invalid move changes the board and every terminal state is clear. Task 2 is ready to commit!